

Leukemia — Organomegaly Overview

Leukemia	Lymphadenopathy	Splenomegaly	Hepatomegaly	Mechanism
CML Chronic Myeloid	✗	✓✓✓ Massive	✓ Mild–moderate	Myeloid cells overflow marrow → spleen becomes backup marrow (extramedullary haematopoiesis). Myeloid cells do not home to lymph nodes.
CLL Chronic Lymphocytic	✓✓✓ Prominent	✓✓ Moderate	✓ Mild–moderate	Mature B cells naturally home to lymph nodes and spleen (their normal organs). Slow accumulation over years → both enlarged.
ALL Acute Lymphoblastic	✓✓ Moderate	✓✓ Moderate	✓✓ Moderate	Aggressive lymphoblasts are highly migratory and infiltrate multiple organs early. Liver sinusoids are easily infiltrated by circulating blasts.
AML Acute Myeloid	✗ (rare)	✓ Mild	✓ Mild	Same myeloid logic as CML but acute = less time to accumulate. Some extramedullary haematopoiesis but far less dramatic than CML.

Key principle: the pattern of organ enlargement reflects where each cell type normally lives. Myeloid cells → marrow/spleen overflow. Lymphoid cells → lymph nodes + spleen infiltration. Liver is a passive bystander in all — its sinusoids are too easy to infiltrate.